IRSwaption 평가 설명서

1. Black Formula for Swaption

Swaption의 기초자산을 F(=forward swap rate)라고 하자.

$$F(0, \mathrm{T_{start}}, \mathrm{T_{end}}) = \frac{\mathrm{P}(0, \mathrm{T_{start}}) - P(0, T_{end})}{\sum_{i=1}^{N} \Delta(T_{i-1}, T_i) P(0, T_i)}$$

$$Payer\ \ Swaption\ \ Price$$
$$= N \times Annuity \times [f_0(T_{start}, T_{end}) N(d_1) - X N(d_2)]$$

$$\mathrm{N(y)} = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{y} e^{-\frac{z^2}{2}} dz$$

$$\mathrm{d_1} = \frac{\left(\ln\left(\frac{f_0(T_{start}, T_{end})}{X}\right) + \frac{1}{2} v^2 T\right)}{v\sqrt{T}} \ \ , \mathrm{d_2} = d_1 - v\sqrt{T}$$

2. Bermudan Swaption

HW 1F FDM을 통해 계산

HW 1F PDE

$$\mathrm{u_t} = \kappa x u_x - \frac{1}{2}\sigma^2 u_{xx} + [\alpha(t) + x]u$$

$$\alpha(\mathrm{t}) = -\frac{\delta P(0,t)}{\delta t} + \frac{\sigma^2}{2\kappa^2}(1 - e^{-\kappa t})^2$$

IFDM

$$\frac{\mathrm{u_i^{k+1}} - \mathrm{u_i^k}}{\mathrm{h_t}} = \kappa \mathrm{x} \cdot \frac{\left(\mathrm{u_{i+1}^k} - \mathrm{u_{i-1}^k}\right)}{2\mathrm{h_x}} - \frac{1}{2}\sigma^2 \frac{\left(\mathrm{u_{i+1}^k} + \mathrm{u_{i-1}^k} - 2\mathrm{u_i^k}\right)}{\mathrm{h_x^2}} + [\alpha(t_{k+1}) + \mathrm{x}_i]\mathrm{u_i^{k+1}}$$

$$\mathrm{u_i^{k+1}} - \mathrm{u_i^k} = \mathrm{h_t}\kappa \mathrm{x} \cdot \frac{\left(\mathrm{u_{i+1}^k} - \mathrm{u_{i-1}^k}\right)}{2\mathrm{h_x}} - \frac{\mathrm{h_t}}{2}\sigma^2 \frac{\left(\mathrm{u_{i+1}^k} + \mathrm{u_{i-1}^k} - 2\mathrm{u_i^k}\right)}{\mathrm{h_x^2}}$$
$$+ \mathrm{h_t}[\alpha(t_{k+1}) + \mathrm{x}_i]\mathrm{u_i^{k+1}}$$

$$\mathrm{u}_{\mathrm{i}}^{\mathrm{k}+1}(1-h_t[\alpha(t_{k+1})+\mathrm{x}_i])$$
$$=\left[-\frac{\mathrm{h_t}\kappa x_i}{2h_x}-\frac{h_t\sigma^2}{2h_x^2}\right]u_{i-1}^k+\left[1+\frac{\mathrm{h_t}\sigma^2}{\mathrm{h_x^2}}\right]u_i^k+\left[\frac{\mathrm{h_t}\kappa x_i}{2h_x}-\frac{h_t\sigma^2}{2h_x^2}\right]u_{i+1}^k$$
$$\mathrm{u}_{\mathrm{i}}^{\mathrm{k}+1}W_i=A_iu_{i-1}^k+Bu_i^k+C_iu_{i+1}^k$$

$$\mathrm{W_i}=(1-h_t[\alpha(t_{k+1})+\mathrm{x}_i])$$
$$A_i=\left[-\frac{\mathrm{h_t}\kappa x_i}{2h_x}-\frac{h_t\sigma^2}{2h_x^2}\right]$$
$$B=\left[1+\frac{\mathrm{h_t}\sigma^2}{\mathrm{h_x^2}}\right]$$
$$C_i=\left[\frac{\mathrm{h_t}\kappa x_i}{2h_x}-\frac{h_t\sigma^2}{2h_x^2}\right]$$

Boundary Condition

$$u_{-1}=2u_0-u_1$$
$$u_{N+1}=2u_N-u_{N-1}$$
$$\mathrm{u}_0^{\mathrm{k}+1}W_0=A_0u_{-1}^k+Bu_0^k+C_iu_1^k=A_0(2u_0^k-u_1^k)+Bu_0^k+C_0u_1^k$$
$$\mathrm{u}_0^{\mathrm{k}+1}W_0=u_0^k(2A_0+B)+u_1^k(C_0-A_0)$$
$$\mathrm{u}_{\mathrm{N}}^{\mathrm{k}+1}W_N=A_Nu_{N-1}^k+Bu_N^k+C_iu_{N+1}^k=A_Nu_{N-1}^k+Bu_N^k+C_N\left(2u_N^k-u_{N-1}^k\right)$$
$$\mathrm{u}_{\mathrm{N}}^{\mathrm{k}+1}W_N=\mathrm{u}_{\mathrm{N}-1}^k(A_N-C_N)+u_N^k(B+2C_N)$$

$$\begin{bmatrix} 2A_0+B & C_0-A_0 & 0 & 0 & \dots & \dots & 0 & 0 & 0 \\ A_1 & B & C_1 & 0 & \dots & \dots & 0 & 0 & 0 \\ 0 & A_2 & B & C_2 & \dots & \dots & 0 & 0 & 0 \\ 0 & 0 & A_3 & B & \dots & \dots & 0 & 0 & 0 \\ \dots & \dots & \dots & \dots & \dots & \dots & 0 & 0 & 0 \\ \dots & \dots & \dots & \dots & \dots & \dots & 0 & 0 & 0 \\ \dots & \dots & \dots & 0 & 0 & A_{N-2} & B & C_{N-2} & 0 \\ \dots & \dots & \dots & 0 & 0 & 0 & A_{N-1} & B & C_{N-1} \\ \dots & \dots & \dots & 0 & 0 & 0 & 0 & A_N-C_N & B+2C_N \end{bmatrix} \begin{bmatrix} u_0^k \\ u_1^k \\ u_2^k \\ u_3^k \\ \dots \\ \dots \\ u_{N-2}^k \\ u_{N-1}^k \\ u_N^k \end{bmatrix}$$

$$= \begin{bmatrix} W_0 u_0^{k+1} \\ W_1 u_1^{k+1} \\ W_2 u_2^{k+1} \\ W_3 u_3^{k+1} \\ \dots \\ \dots \\ W_{N-2} u_{N-2}^{k+1} \\ W_{N-1} u_{N-1}^{k+1} \\ W_N u_N^{k+1} \end{bmatrix}$$

HW Forward Discount Factor

$$\mathrm{P}_{\mathrm{HW}}^{x_i}(\mathrm{t},\mathrm{T}) = \frac{P(T)}{P(t)} \exp\left(-x_i \times B(t,T) + \frac{1}{2} QVTerm(t,T)\right)$$

$$B(t,T) = \frac{\left(1-\exp\left(-\kappa(T-t)\right)\right)}{\kappa}$$

$$QVTerm(t,T) = \int_0^t \sigma^2 [B(s,t)^2 - B(s,T)^2] ds$$

$$T \rightarrow Swaption\ \ SwapMaturity\ \ T$$

HW Forward Swap Rate

$$\mathrm{FSR}_{\mathrm{HW}}^{x_i}(\mathrm{t}_{\mathrm{timenode}}, \mathrm{t}_{\mathrm{forwardstart}}, \mathrm{T}_{\mathrm{swapmat}}) = \frac{P_{HW}^{x_i}(t_{timenode}, t_{forwardstart}) - P_{HW}^{x_i}(t_{timenode}, T_N)}{\sum_1^N \left[\Delta T_i \times P_{HW}^{x_i}(t_{timenode}, T_k)\right]}$$

$$* \ \ \mathrm{t}_{\mathrm{forwardstart}} = \mathrm{Forward\ \ Swap\ \ Start\ \ T}$$

$$* \ \ \mathrm{T}_1 = \mathrm{First\ \ Swap\ \ Coupon\ \ T}$$

$$* \ \ \mathrm{T}_{\mathrm{N}} = T_{\mathrm{swapmat}}$$

HW Swaption Value

- Swaption Value on node $\mathrm{x_i}$

$$\mathrm{U}(\mathrm{t}_{\mathrm{timenode}}, \mathrm{x}_{\mathrm{i}}) = max\left(\mathrm{FSR}_{\mathrm{HW}}^{x_i}\left(\mathrm{t}_{\mathrm{timenode}}, \mathrm{t}_{\mathrm{forwardstart}}, \mathrm{T}_{\mathrm{swapmat}}\right) - Strike, 0\right) \sum_{1}^{N} \left[\Delta T_i \times P_{HW}^{x_i}(t_{timenode}, T_i)\right]$$

그리드 세팅 전에 미리 만들어야 할 변수들 3D Array

$$\{B(t, T_k) | k = 0 \ \ to \ \ N\} \rightarrow NOption \times Time \ \ Greed\text{개수만큼}$$

$$\{QVTerm(t, T_k) | k = 0 \ \ to \ \ N\} \rightarrow NOption \times Time \ \ Greed\text{개수만큼}$$

$$\{P(0, T_k) | k = 0 \ \ to \ \ N\} \rightarrow NOption \times Time \ \ Greed\text{개수만큼}$$